

Math 231A HW 7

Zih-Yu Hsieh

November 13, 2025

1 D

Problem 1

Etingof Lecture Notes Exercise 15.5:
Prove Lemma 15.4:

Lemma

The quotient $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is abelian; moreover, if $I \subseteq \mathfrak{g}$ is an ideal such that $\mathfrak{g}/I$ is abelian, then $I \supseteq [\mathfrak{g}, \mathfrak{g}]$.

Solution: First, to show that the Lie bracket on the quotient of a Lie ideal is well-defined, given any $x, x', y, y' \in \mathfrak{g}$ such that $\bar{x} = \bar{x}'$ and $\bar{y} = \bar{y}'$ under $\mathfrak{g}/I$, where I is a Lie ideal. Then, this indicates that $x - x', y - y' \in I$. Hence, by the property that I is a Lie ideal, we have the following:

$$[x, y] - [x, y'] = [x, y - y'] \in I \implies \overline{[x, y]} = \overline{[x, y']} \quad (1.1)$$

$$[x, y'] - [x', y'] = [x - x', y'] \in I \implies \overline{[x, y']} = \overline{[x', y']} \quad (1.2)$$

Hence, this indicates that the quotient $\overline{[x, y]} = \overline{[x', y']}$, showing that the Lie bracket is well-defined on the level of quotient. Hence, it's well-defined to set $[\bar{x}, \bar{y}] := \overline{[x, y]}$.

Now, given any $\bar{x}, \bar{y} \in \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$, since $[\bar{x}, \bar{y}] = \overline{[x, y]} = 0$ (since $[x, y] \in [\mathfrak{g}, \mathfrak{g}]$), then $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is abelian.

Finally, given that $\mathfrak{g}/I$ is abelian for a Lie ideal I , then given any $x, y \in \mathfrak{g}$, we have that $[\bar{x}, \bar{y}] = \overline{[x, y]} = 0$, showing that $[x, y] \in I$. Hence, since all the generators of $[\mathfrak{g}, \mathfrak{g}]$ is contained in I , we have $[\mathfrak{g}, \mathfrak{g}] \subseteq I$.

2 D

Problem 2

Etingof Lecture Notes Exercise 15.7:

- (i) Prove that if G is a connected Lie group with Lie algebra $\mathfrak{g}$ then the group commutant $[G, G]$ (the subgroup of G generated by elements $ghg^{-1}h^{-1}$, $g, h \in G$) is a Lie subgroup of G with Lie algebra $[\mathfrak{g}, \mathfrak{g}]$.
- (ii) Let $\tilde{G} = \mathbb{R} \times H$, where H is the *Heisenberg Group* of real matrices of the form

$$M(a, b, c) := \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}, \quad a, b, c \in \mathbb{R} \quad (2.1)$$

Let $\Gamma \cong \mathbb{Z}^2 \subset \tilde{G}$ be the (closed) central subgroup generated by the pairs $(1, M(0, 0, 0) = \text{id})$ and $(\sqrt{2}, M(0, 1, 0))$. Let $G = \tilde{G}/\Gamma$. Show that $[G, G]$ is not closed in G (although by (i) it is a Lie subgroup).

- (iii) Does $[G, G]$ have to be closed in G if G is simply-connected? (Consider $\text{Hom}(G, \mathbb{R})$ and apply the second fundamental theorem of Lie theory).

Solution:

- (i) WLOG, we'll take the universal cover $\tilde{G}$ instead of G , since the covering map $p : \tilde{G} \rightarrow G$ is a local diffeomorphism satisfying $p([\tilde{G}, \tilde{G}]) = [G, G]$ ($[\tilde{G}, \tilde{G}]$ is a subgroup generated by the paths of the form $[x(t)y(t)x(t)^{-1}y(t)^{-1}]$, which gets mapped down to $x(1)y(1)x(1)^{-1}y(1)^{-1} \in [G, G]$. Since G is connected, $x(1), y(1)$ can be arbitrary, so the generators of $[G, G]$ is contained in $p([\tilde{G}, \tilde{G}])$, hence $[G, G] \subseteq p([\tilde{G}, \tilde{G}])$; the other direction is given by the fact that every generator of $[\tilde{G}, \tilde{G}]$ is mapped into $[G, G]$, hence $p([\tilde{G}, \tilde{G}]) \subseteq [G, G]$. Then, if proving that $[\tilde{G}, \tilde{G}]$ is a submanifold corresponding to $[\mathfrak{g}, \mathfrak{g}]$, it follows from the covering map that $[G, G] = p([\tilde{G}, \tilde{G}])$ is also a submanifold corresponding to $[\mathfrak{g}, \mathfrak{g}]$ (since the evenly covered neighborhood of $1 \in G$ and $[1] \in \tilde{G}$ shows that their tangent space must be isomorphic via the differential of covering map p_* , hence also isomorphic as Lie Algebra).

To comprehend this, on the Lie algebra level, we have the following „exact sequence“ (where π is the quotient map):

$$0 \longrightarrow [\mathfrak{g}, \mathfrak{g}] \hookrightarrow \mathfrak{g} \xrightarrow{\pi} \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] \longrightarrow 0$$

Using Fundamental Theorem of Lie Theory (equivalence of categories between simply-connected Lie groups and Lie algebras), take the simply-connected Lie group of $\mathfrak{g}$ (which $\tilde{G}$ satisfies this), and $\tilde{K}$ of $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$. The Second Fundamental Theorem of Lie Theory states the bijection between $\text{Hom}(\tilde{G}, \tilde{K}) \simeq \text{Hom}(\mathfrak{g}, \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}])$ by sending the Lie group homomorphism $\varphi \mapsto \varphi_*$, hence for the quotient map $\pi : \mathfrak{g} \rightarrow \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$, there exists a unique Lie group homomorphism $\varphi : \tilde{G} \rightarrow \tilde{K}$, such that $\varphi_* = \pi$. Which, let $H := \ker(\varphi)$ be a Lie subgroup of $\tilde{G}$, it forms the following correspondance of exact sequences:

$$\begin{array}{ccccccc} 0 & \longrightarrow & H & \hookrightarrow & \tilde{G} & \xrightarrow{\varphi} & \tilde{K} \longrightarrow 0 \\ & & & & \uparrow \exp_{\tilde{G}} & & \uparrow \exp_{\tilde{K}} \\ 0 & \longrightarrow & [\mathfrak{g}, \mathfrak{g}] & \hookrightarrow & \mathfrak{g} & \xrightarrow{\pi=\varphi_*} & \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] \longrightarrow 0 \end{array}$$

Which, this correspondance also shows that $\text{Lie}(H) = [\mathfrak{g}, \mathfrak{g}]$: Given any $x \in [\mathfrak{g}, \mathfrak{g}]$, since they satisfy the following:

$$\varphi \circ \exp_{\tilde{G}}(x) = \exp_{\tilde{K}} \circ \pi(x) = \exp_{\tilde{K}}(0) = 1_{\tilde{K}} \quad (2.2)$$

then $\exp_{\tilde{G}}(x) \in \ker(\varphi) = H$, hence $[\mathfrak{g}, \mathfrak{g}] \subseteq \text{Lie}(H)$; on the other hand, take $H \cap U$ (where $U \subseteq \tilde{G}$ is the domain of $\log_{\tilde{G}}$ function), then any $h \in H \cap U$ satisfies the following (recall that $\log$ is inverse of $\exp$ on some open subset):

$$\pi \circ \log_{\tilde{G}}(h) = \log_{\tilde{K}} \circ \varphi(h) = \log_{\tilde{K}}(1_{\tilde{K}}) = 0 \quad (2.3)$$

Hence, showing that the tangent space $\text{Lie}(H) \subseteq \ker(\pi) = [\mathfrak{g}, \mathfrak{g}]$. So, the full correspondance is as follow:

$$\begin{array}{ccccccc} 0 & \longrightarrow & H & \hookrightarrow & \tilde{G} & \xrightarrow{\varphi} & \tilde{K} \longrightarrow 0 \\ & & \uparrow \exp_H & & \uparrow \exp_{\tilde{G}} & & \uparrow \exp_{\tilde{K}} \\ 0 & \longrightarrow & [\mathfrak{g}, \mathfrak{g}] & \hookrightarrow & \mathfrak{g} & \xrightarrow{\pi=\varphi_*} & \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] \longrightarrow 0 \end{array}$$

Now, since $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is abelian Lie algebra, then it indicates that $\tilde{K}$ is an abelian Lie group, in particular $[\tilde{G}, \tilde{G}] \subseteq \ker(\varphi) = H$ (since image of φ is abelian).

For the reverse direction, given any $x, y \in \mathfrak{g}$, we have the curve $e^{tx}e^ye^{-tx}e^{-y} : \mathbb{R} \rightarrow G$ satisfies the following:

$$\begin{aligned} \frac{\log(e^{tx}e^ye^{-tx}e^{-y})}{t} &= \frac{\log(\text{Ad}_{e^{tx}}(e^y)e^{-y})}{t} = \frac{\log(\exp(\text{Ad}_{e^{tx}*}y)e^{-y})}{t} \\ &= \frac{\text{Ad}_{e^{tx}*} \cdot y + (-y) + [\text{Ad}_{e^{tx}*} \cdot y, -y] + \dots}{t} \\ &= \frac{\exp(\text{ad}(tx))(y) - y + [\exp(\text{ad}(tx))(y), -y] + \dots}{t} \\ &= \frac{\sum_{n=0}^{\infty} \frac{t^n}{n!} \text{ad}(x)^n(y) - y - \sum_{n=0}^{\infty} \frac{t^n}{n!} [\text{ad}(x)^n(y), y] + \dots}{t} \\ &= \frac{y + t \text{ad}(x)(y) + \sum_{n=2}^{\infty} \frac{t^n}{n!} \text{ad}(x)^n(y) - y - [y, y] - \sum_{n=1}^{\infty} \frac{t^n}{n!} [\text{ad}(x)^n(y), y] + \dots}{t} \\ &= \text{ad}(x)(y) + \left(\sum_{n=2}^{\infty} \frac{t^{n-1}}{n!} \text{ad}(x)^n(y) - \sum_{n=1}^{\infty} \frac{t^{n-1}}{n!} [\text{ad}(x)^n(y), y] + \dots \right) \end{aligned} \quad (2.4)$$

Which, the eventual ... using BCH formula can be seen that it involves higher order terms of $\text{Ad}_{e^{tx}*}(y)$ bracketing with y , hence expanding $\text{Ad}_{e^{tx}*} = \exp(\text{ad}(tx))$, the zeroth order term cancels (by bracketing with y itself), while every other term has degree of t being greater than 1. Hence, take $\lim_{t \rightarrow 0} \frac{\log(e^{tx}e^ye^{-tx}e^{-y})}{t}$, the above formula provides $\text{ad}(x)(y) = [x, y]$.

Since the curve $e^{tx}e^ye^{-tx}e^{-y} \subset [\tilde{G}, \tilde{G}]$, one has $\exp([x, y]) \in [\tilde{G}, \tilde{G}]$; also, since all $[x, y]$ are generators of $[\mathfrak{g}, \mathfrak{g}]$, its exponentiation should generate a subgroup containing in $[\tilde{G}, \tilde{G}]$. Finally, with H being generated by exponentiation of $[\mathfrak{g}, \mathfrak{g}]$, we conclude that $H \subseteq [\tilde{G}, \tilde{G}]$. Hence, we must have $[\tilde{G}, \tilde{G}] = H$, showing $[\tilde{G}, \tilde{G}]$ is in fact a Lie subgroup.

With the covering map claim, $p([\tilde{G}, \tilde{G}]) = [G, G]$ is also a Lie subgroup of G (since it's an immersed submanifold).

- (ii) Since the projection $\pi : \tilde{G} \rightarrow \tilde{G}/\Gamma = G$ defines the topology of G through quotient topology, to show that $[G, G]$ is not closed, it suffices to compute the preimage $\pi^{-1}([G, G])$ being not closed (since under quotient topology, a subset V is closed iff its preimage is closed).

First, given any $g, h \in \tilde{G}$, we have the commutator $\pi(ghg^{-1}h^{-1})$ being a commutator of $\pi(g), \pi(h)$ instead, so π sends generators of $[\tilde{G}, \tilde{G}]$ to $[G, G]$, showing $\pi([\tilde{G}, \tilde{G}]) \subseteq [G, G]$. Also, since the projection π is surjective, then for every $g', h' \in G$, there exists $g, h \in \tilde{G}$ in

the fibre of g', h' respectively. Which, $g'h'g'^{-1}h'^{-1} = \pi(ghg^{-1}h^{-1})$, showing generators of $[G, G]$ is contained in $\pi([\tilde{G}, \tilde{G}])$, hence $[G, G] \subseteq \pi([\tilde{G}, \tilde{G}])$. So, $[G, G] = \pi([G, G])$.

Hence, for all $g \in \pi^{-1}([G, G])$, there is an $h \in [\tilde{G}, \tilde{G}]$, such that $\pi(g) = \pi(h)$, or $\pi(h^{-1}g) = 1$, hence $h^{-1}g \in \Gamma$, showing $g \in h\Gamma \subset [\tilde{G}, \tilde{G}]\Gamma$, showing $\pi^{-1}([G, G]) \subseteq [\tilde{G}, \tilde{G}]\Gamma$. (which, because $[\tilde{G}, \tilde{G}]$ is normal and Γ is central, $[\tilde{G}, \tilde{G}]\Gamma$ is a well-defined subgroup). Conversely, all element in $[\tilde{G}, \tilde{G}]\Gamma$ is of the form ab where $a \in [\tilde{G}, \tilde{G}]$ and $b \in \Gamma$, so $\pi(ab) = \pi(a)\pi(b) = \pi(a)$, where $\pi(a) \in [G, G]$, showing that $\pi([\tilde{G}, \tilde{G}]\Gamma) \subseteq [G, G]$, hence $[\tilde{G}, \tilde{G}]\Gamma \subseteq \pi^{-1}([G, G])$. So, $\pi^{-1}([G, G]) = [\tilde{G}, \tilde{G}]\Gamma$.

Now, based on the claim, it suffices to compute that $[\tilde{G}, \tilde{G}]\Gamma$ is not closed. Given any $(x, M(a, b, c)), (y, M(d, e, f)) \in \tilde{G}$, their commutator is given as follow:

$$\begin{aligned}
 & [(x, M(a, b, c)), (y, M(d, e, f))] = (x + y - x - y, [M(a, b, c), M(d, e, f)]) \\
 & = \left(0, \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{pmatrix}^{-1} \right) \\
 & = \left(0, \begin{pmatrix} 1 & a & af+b+e \\ 0 & 1 & c+f \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -a & -b+ac \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -d & -e+df \\ 0 & 1 & -f \\ 0 & 0 & 1 \end{pmatrix} \right) \\
 & = \left(0, \begin{pmatrix} 1 & 0 & (-b+ac) + a(-c) + (af+b+e) \\ 0 & 1 & -c + (c+f) \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -d & -e+df \\ 0 & 1 & -f \\ 0 & 0 & 1 \end{pmatrix} \right) \quad (2.5) \\
 & = \left(0, \begin{pmatrix} 1 & 0 & af+e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -d & -e+df \\ 0 & 1 & -f \\ 0 & 0 & 1 \end{pmatrix} \right) \\
 & = \left(0, \begin{pmatrix} 1 & 0 & af+df \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right)
 \end{aligned}$$

Which, this shows that generators is contained in the subgroup $K = \{(0, M(0, x, 0)) \mid x \in \mathbb{R}\}$, or $[\tilde{G}, \tilde{G}] \subseteq K$; also, take $a = x, f = 1$, and $d = 0$, then the above commutator generates $(0, M(0, x, 0))$, hence we have $[\tilde{G}, \tilde{G}] = K$.

Finally, since $\Gamma = \langle (1, M(0, 0, 0)), (\sqrt{2}, M(0, 1, 0)) \rangle$ is central, everything has the structure of $(1, M(0, 0, 0))^n (\sqrt{2}, M(0, 1, 0))^m = (n + m\sqrt{2}, M(0, m, 0))$ based on matrix computation (where $n, m \in \mathbb{Z}$). So, with $[\tilde{G}, \tilde{G}] = K = \{(0, M(0, x, 0)) \mid x \in \mathbb{R}\}$, we have the following:

$$(0, M(0, x, 0))(n + m\sqrt{2}, M(0, m, 0)) = (n + m\sqrt{2}, M(0, x + m, 0)) \quad (2.6)$$

Hence, this indicates that $[\tilde{G}, \tilde{G}]\Gamma = \{(n + m\sqrt{2}, M(0, x + m, 0)) \mid n, m \in \mathbb{Z}, x \in \mathbb{R}\}$. But, notice that this is not closed, since one can choose the subset where $x = -m$, so the corresponding subset is $\{(n + m\sqrt{2}, M(0, 0, 0)) \mid n, m \in \mathbb{Z}\}$. Since the subset $\{n + m\sqrt{2} \mid n, m \in \mathbb{Z}\} \subset \mathbb{R}$ is dense (but not spanning the whole $\mathbb{R}$), we have for instance $(\sqrt{3}, M(0, 0, 0))$ being a limit point but not in the set (since if $\sqrt{3} = n + m\sqrt{2}$, one has $2 = n^2 + 2m^2 + 2nm\sqrt{2}$; yet with $\sqrt{2}$ being irrational, we must have $nm = 0$, showing $n = 0$ or $m = 0$. But, $n = 0$ implies $\sqrt{3} = m\sqrt{2}$, or $3 = 2m^2$ is a contradiction by parity; similarly, $m = 0$ implies $\sqrt{3} = n$ is also a contradiction by rationality). So, $(\sqrt{3}, M(0, 0, 0)) \notin [\tilde{G}, \tilde{G}]\Gamma$, but is a limit point, showing that $[\tilde{G}, \tilde{G}]\Gamma = \pi^{-1}([G, G])$ is not closed, so is $[G, G]$.

- (iii) If G is simply-connected, by the Second Fundamental Theorem of Lie Theory, the map $\text{Hom}(G, \mathbb{R}) \rightarrow \text{Hom}(\mathfrak{g}, \text{Lie}(\mathbb{R}))$ by sending the Lie group homomorphism $\varphi \mapsto \varphi_*$ defines a bijection.

Since as Lie group $\mathbb{R}$ is abelian (under addition), hence every Lie group homomorphism $\varphi : G \rightarrow \mathbb{R}$ satisfies $[G, G] \subseteq \ker(\varphi)$, hence on the tangent space level, we have $[\mathfrak{g}, \mathfrak{g}] = \text{Lie}([G, G]) \subseteq \text{Lie}(\ker(\varphi)) = \ker(\varphi_*)$ (Note: such correspondance is true, because of the equality $\varphi \circ \exp_G = \exp_{\mathbb{R}} \circ \varphi_*$, hence $x \in \ker(\varphi_*)$ iff $\exp_{\mathbb{R}} \circ \varphi_*(x) = \exp_{\mathbb{R}}(0) = 0_{\mathbb{R}}$ iff $\varphi \circ \exp_G(x) = 0_{\mathbb{R}}$ iff $\exp_G(x) \in \ker(\varphi)$, showing the tangent space of $\ker(\varphi)$ is $\ker(\varphi_*)$). This also indicates that $[\mathfrak{g}, \mathfrak{g}] \subseteq \bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} \ker(\varphi_*)$.

Now, we claim the inclusion is in fact equality: Given any $x \notin [\mathfrak{g}, \mathfrak{g}]$, choose a basis $\ell_1, \dots, \ell_k$ for $[\mathfrak{g}, \mathfrak{g}]$, then $\ell_1, \dots, \ell_k, x$ is linearly independent in $\mathfrak{g}$, hence can be extended to a basis of $\mathfrak{g}$, say $\ell_1, \dots, \ell_k, x, f_1, \dots, f_l$ (say $\mathfrak{g}$ is $k + l + 1$ -dimension). Then, define the Linear functional $\varphi_x : \mathfrak{g} \rightarrow \mathbb{R}$ by $\varphi_x(f_i) = \varphi_x(\ell_j) = 0$, while $\varphi_x(x) = 1$. This is in fact a well-defined Lie algebra homomorphism, since given any $a, b \in \mathfrak{g}$, one has $[a, b] \in [\mathfrak{g}, \mathfrak{g}] = \text{span}\{\ell_1, \dots, \ell_k\}$, hence $\varphi_x([a, b]) = 0 = [\varphi_x(a), \varphi_x(b)]$ (recall that $\mathbb{R} = \text{Lie}(\mathbb{R})$, and it's abelian, so every Lie bracket is 0). So, there exists $\varphi_x \in \text{Hom}(\mathfrak{g}, \text{Lie}(\mathbb{R}))$, such that $x \notin \ker(\varphi_x)$, showing that $x \notin \bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} \ker(\varphi_*)$. In other words, one must have $\bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} \ker(\varphi_*) \subseteq [\mathfrak{g}, \mathfrak{g}]$, showing $[\mathfrak{g}, \mathfrak{g}] = \bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} \ker(\varphi_*)$.

As a result, first since each $\varphi \in \text{Hom}(G, \mathbb{R})$ satisfies $[G, G] \subseteq \ker(\varphi)$, so $[G, G] \subseteq \bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} \ker(\varphi)$ as subgroup due to the property of abelianization.

Also, if we consider some open neighborhood of 1, say U such that $\exp_G$ is a diffeomorphism with inverse $\log$ (which U generates G), since each $\ker(\varphi)$ is a closed Lie subgroup generated by $U \cap \exp_G(\ker(\varphi_*)) = U \cap \ker(\varphi)$ (exponentiation of their own tangent space), then if consider $U \cap \left(\bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} \ker(\varphi) \right) = \bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} (U \cap \ker(\varphi)) = \bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} (U \cap \exp_G(\ker(\varphi_*)))$ (which generates $\bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} \ker(\varphi)$ as groups, not manifold yet), since U is where $\exp_G$ becomes a diffeomorphism (in particular a bijection), then $\bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} (U \cap \exp_G(\ker(\varphi_*)))$ must be exponentiated from subsets of $\bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} \ker(\varphi_*) = [\mathfrak{g}, \mathfrak{g}]$ (since then $\log(U \cap \exp_G(\ker(\varphi_*))) \subseteq \ker(\varphi_*)$, so $\log\left(\bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} (U \cap \exp_G(\ker(\varphi_*)))\right) = \bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} \log(U \cap \exp_G(\ker(\varphi_*))) \subseteq \bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} \ker(\varphi_*) = [\mathfrak{g}, \mathfrak{g}]$).

So, $\bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} \ker(\varphi)$ can be generated by exponentiation of subsets of $[\mathfrak{g}, \mathfrak{g}]$, which is contained in $[G, G]$ (which has tangent space $[\mathfrak{g}, \mathfrak{g}]$). Hence, $[G, G] = \bigcap_{\varphi \in \text{Hom}(G, \mathbb{R})} \ker(\varphi)$.

With each $\ker(\varphi)$ being closed Lie subgroup, this indicates that $[G, G]$ as their intersection is closed. So, G being simply-connected implies $[G, G]$ is closed.

3 D

Problem 3

Etingof Lecture Notes Exercise 15.11:
Prove Proposition 15.10:

Proposition

- (i) Any Lie subalgebra or quotient of a solvable Lie algebra is solvable.
- (ii) If $I \subseteq \mathfrak{g}$ is an ideal and $I, \mathfrak{g}/I$ are solvable, then $\mathfrak{g}$ is solvable.

Solution:

- (i) First, given a solvable Lie algebra $\mathfrak{g}$, suppose $\mathfrak{h} \subseteq \mathfrak{g}$ is a subalgebra. Then, we'll prove by induction that the derived series of $\mathfrak{h}$ is always contained in $\mathfrak{g}$'s series.

For $n = 1$, all $x, y \in \mathfrak{h}$ has $[x, y] \in [\mathfrak{g}, \mathfrak{g}]$, so all generators of $[\mathfrak{h}, \mathfrak{h}]$ is contained in $[\mathfrak{g}, \mathfrak{g}]$, showing $D^1(\mathfrak{h}) = [\mathfrak{h}, \mathfrak{h}] \subseteq [\mathfrak{g}, \mathfrak{g}] = D^1(\mathfrak{g})$.

Now, suppose given $n \in \mathbb{N}$, $n - 1 < n$ satisfies $D^{n-1}(\mathfrak{h}) \subseteq D^{n-1}(\mathfrak{g})$. Then, for any $x, y \in D^{n-1}(\mathfrak{h})$, we have $[x, y] \in [D^{n-1}(\mathfrak{g}), D^{n-1}(\mathfrak{g})] = D^n(\mathfrak{g})$, so all generators of $D^n(\mathfrak{h}) = [D^{n-1}(\mathfrak{h}), D^{n-1}(\mathfrak{h})]$ is contained in $D^n(\mathfrak{g})$, showing $D^n(\mathfrak{h}) \subseteq D^n(\mathfrak{g})$. Hence, by induction one concludes that all $D^n(\mathfrak{h}) \subseteq D^n(\mathfrak{g})$.

As a result, since $D^n(\mathfrak{g}) = 0$ for some $n \in \mathbb{N}$ by solvability, then it enforces $D^n(\mathfrak{h}) = 0$, showing $\mathfrak{h}$ is also solvable.

Now, given any Lie ideal $I \subseteq \mathfrak{g}$ and the quotient map $\pi : \mathfrak{g} \twoheadrightarrow \mathfrak{g}/I$ (for simplicity, let $\mathfrak{k} = \mathfrak{g}/I$). Again, we'll prove by induction that the derived series of $\mathfrak{k}$ is the image of $\mathfrak{g}$'s derived series.

For $n = 1$, since for all $\bar{x}, \bar{y} \in \mathfrak{k}$ satisfies $[\bar{x}, \bar{y}] = \overline{[x, y]} = \pi([x, y])$, hence the generators of $D^1(\mathfrak{k}) = [\mathfrak{k}, \mathfrak{k}]$ is contained in $\pi(D^1(\mathfrak{g})) = \pi([\mathfrak{g}, \mathfrak{g}])$, hence $D^1(\mathfrak{k}) \subseteq \pi(D^1(\mathfrak{g}))$. Conversely, it also suggests that the generators $[x, y]$ of $D^1(\mathfrak{g})$ has image contained in $D^1(\mathfrak{k})$, hence $\pi(D^1(\mathfrak{g})) \subseteq D^1(\mathfrak{k})$. This concludes that $\pi(D^1(\mathfrak{g})) = D^1(\mathfrak{k})$.

Now, suppose given $n \in \mathbb{N}$, $n - 1 < n$ satisfies $D^{n-1}(\mathfrak{k}) = \pi(D^{n-1}(\mathfrak{g}))$. Then, for any $\bar{x}, \bar{y} \in D^{n-1}(\mathfrak{k}) = \pi(D^{n-1}(\mathfrak{g}))$ (where they correspond to the image of some $x, y \in \pi(D^{n-1}(\mathfrak{g}))$), one has $[\bar{x}, \bar{y}] = \overline{[x, y]}$, where $[x, y] \in [D^{n-1}(\mathfrak{g}), D^{n-1}(\mathfrak{g})] = D^n(\mathfrak{g})$. This shows that $[\bar{x}, \bar{y}] \in \pi(D^n(\mathfrak{g}))$, hence all generators of $D^n(\mathfrak{k}) = [D^{n-1}(\mathfrak{k}), D^{n-1}(\mathfrak{k})]$ is contained in $\pi(D^n(\mathfrak{g}))$, therefore $D^n(\mathfrak{k}) \subseteq \pi(D^n(\mathfrak{g}))$. Conversely, this also shows that generators of $D^n(\mathfrak{g})$ (all $[x, y]$ with $x, y \in D^{n-1}(\mathfrak{g})$) satisfies $\pi([x, y]) \in D^n(\mathfrak{k})$, hence $\pi(D^n(\mathfrak{g})) \subseteq D^n(\mathfrak{k})$. This concludes that $\pi(D^n(\mathfrak{g})) = D^n(\mathfrak{k})$. So, by induction one concludes that all $D^n(\mathfrak{k}) = \pi(D^n(\mathfrak{g}))$.

As a result, since $D^n(\mathfrak{g}) = 0$ for some $n \in \mathbb{N}$ by solvability, then it also enforces $D^n(\mathfrak{k}) = 0$ (since $\pi(D^n(\mathfrak{g})) = 0$), showing $\mathfrak{k}$ is also solvable.

- (ii) Now, suppose that both I and $\mathfrak{g}/I = \mathfrak{k}$ are solvable (with quotient map $\pi : \mathfrak{g} \twoheadrightarrow \mathfrak{g}/I$). The goal is to show the derived series eventually is contained in I , which we'll heavily utilize the statement in (i).

Since in (i) we've proven that $D^n(\mathfrak{k}) = \pi(D^n(\mathfrak{g}))$ for all $n \in \mathbb{N}$, hence by solvability of $\mathfrak{k}$, there exists $n \in \mathbb{N}$, where $\pi(D^n(\mathfrak{g})) = D^n(\mathfrak{k}) = 0$. This implies that $D^n(\mathfrak{g}) \subseteq I$. However, since $D^n(\mathfrak{g})$ is now a subalgebra of I , with I being solvable, the statement in (i) implies $D^n(\mathfrak{g})$ is also solvable, hence for some $k \in \mathbb{N}$, $D^k(D^n(\mathfrak{g})) = 0$.

Finally, it suffices to realize that $D^k(D^n(\mathfrak{g})) = D^{n+1}(\mathfrak{g})$:

Since $D^1(D^n(\mathfrak{g})) = [D^n(\mathfrak{g}), D^n(\mathfrak{g})] = D^{n+1}(\mathfrak{g})$, and by induction if $k-1 < k$ satisfies $D^{k-1}(D^n(\mathfrak{g})) = D^{n+k-1}(\mathfrak{g})$, then $D^k(D^n(\mathfrak{g})) = [D^{k-1}(D^n(\mathfrak{g})), D^{k-1}(D^n(\mathfrak{g}))] = [D^{n+k-1}(\mathfrak{g}), D^{n+k-1}(\mathfrak{g})] = D^{n+k}(\mathfrak{g})$.

So, we have $0 = D^k(D^n(\mathfrak{g})) = D^{n+k}(\mathfrak{g})$, showing $\mathfrak{g}$ is solvable.

4 D

Problem 4

Etingof Lecture Notes Exercise 15.16:

Prove Proposition 15.15:

Proposition

Any Lie subalgebra or quotient of a nilpotent Lie algebra is nilpotent.

Solution: First, suppose $\mathfrak{g}$ is a nilpotent Lie algebra, with $\mathfrak{h} \subseteq \mathfrak{g}$ being a subalgebra. To show the nilpotency of $\mathfrak{h}$, it suffices to show its Lower Central Series is always contained in $\mathfrak{g}$'s Lower Central Series.

For $n = 1$, it's clear that $D_1(\mathfrak{h}) = [\mathfrak{h}, \mathfrak{h}] \subseteq [\mathfrak{g}, \mathfrak{g}] = D_1(\mathfrak{g})$ (for proof cf. **Problem 3 part (i)**).

Now, suppose for given $n \in \mathbb{N}$, it satisfies $D_{n-1}(\mathfrak{h}) \subseteq D_{n-1}(\mathfrak{g})$. Then for $D_n(\mathfrak{h}) = [D_{n-1}(\mathfrak{h}), \mathfrak{h}]$, since all its generators are of the form $[x, y]$, where $x \in D_{n-1}(\mathfrak{h}) \subseteq D_{n-1}(\mathfrak{g})$ and $y \in \mathfrak{h} \subseteq \mathfrak{g}$, so $[x, y] \in [D_{n-1}(\mathfrak{g}), \mathfrak{g}] = D_n(\mathfrak{g})$. Hence, all generators of $D_n(\mathfrak{h})$ is contained in $D_n(\mathfrak{g})$, showing $D_n(\mathfrak{h}) \subseteq D_n(\mathfrak{g})$. Hence, by induction all $n \in \mathbb{N}$ satisfies $D_n(\mathfrak{h}) \subseteq D_n(\mathfrak{g})$.

As a result, since $\mathfrak{g}$ is nilpotent, there exists $k \in \mathbb{N}$, where $D_k(\mathfrak{g}) = 0$, and it enforces $D_k(\mathfrak{h}) = 0$ also, so $\mathfrak{h}$ is nilpotent.

Now, given an ideal $I \subseteq \mathfrak{g}$, and the quotient map $\pi : \mathfrak{g} \twoheadrightarrow \mathfrak{g}/I$ (for simplicity again, denote $\mathfrak{g}/I = \mathfrak{k}$). We'll show its Lower Central Series is always contained in the image of $\mathfrak{g}$'s Lower Central Series.

For $n = 1$, it's also clear that $D_1(\mathfrak{k}) = D^1(\mathfrak{k}) \subseteq \pi(D^1(\mathfrak{g})) = \pi(D_1(\mathfrak{g}))$ (again, cf. **Problem 3 part (i)**).

Now, suppose for given $n \in \mathbb{N}$, it satisfies $D_{n-1}(\mathfrak{k}) \subseteq \pi(D_{n-1}(\mathfrak{g}))$, then for $D_n(\mathfrak{k}) = [D_{n-1}(\mathfrak{k}), \mathfrak{k}]$, the generators are of the form $[\bar{x}, \bar{y}]$, where $\bar{x} \in D_{n-1}(\mathfrak{k}) \subseteq \pi(D_{n-1}(\mathfrak{g}))$ (say corresponds to the image of $x \in D_{n-1}(\mathfrak{g})$), and $\bar{y} \in \mathfrak{k}$ (say corresponds to the image of $y \in \mathfrak{g}$). Then, since $[\bar{x}, \bar{y}] = \overline{[x, y]}$, where $[x, y] \in [D_{n-1}(\mathfrak{g}), \mathfrak{g}] = D_n(\mathfrak{g})$, showing that $[\bar{x}, \bar{y}] \in \pi(D_n(\mathfrak{g}))$. Hence, $D_n(\mathfrak{k}) \subseteq \pi(D_n(\mathfrak{g}))$ (since all generators satisfy this relation). Therefore, by induction every $n \in \mathbb{N}$ satisfies $D_n(\mathfrak{k}) \subseteq \pi(D_n(\mathfrak{g}))$.

As a result, since $\mathfrak{g}$ is nilpotent, there exists $k \in \mathbb{N}$ where $D_k(\mathfrak{g}) = 0$, which enforces $D_k(\mathfrak{k}) = 0$ also (since $\pi(D_k(\mathfrak{g})) = 0$). So, $\mathfrak{k}$ is also nilpotent.